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Basic Power Semiconductor Devices

Power Diode, SCR, DIAC, TRIAC, MOSFET, and GaN

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Outline

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Power Diode

SCR (Thyristor)

DIAC

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Quadrant Switch & Combinations

Comparison and Summary

What is Power Electronics? ⚡

Definition

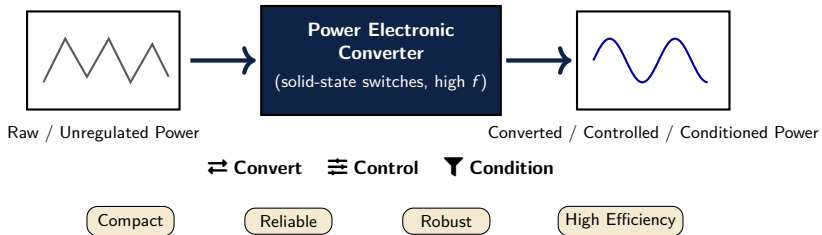
Power Electronics is the branch of electrical engineering and technology concerned with converting, controlling, and conditioning electrical power using solid-state devices as switches – generally operated at high frequency – to achieve compact, reliable, robust, and high-efficiency energy conversion.

Three core actions:

- **Convert** – change the form of electrical power ($AC \leftrightarrow DC$, one voltage/frequency to another)
- **Control** – regulate voltage, current, or power flow to the load
- **Condition** – shape and clean up the power quality delivered

Four design goals: Compact ■ Reliable ■ Robust ■ High Efficiency

Power Electronics – At a Glance



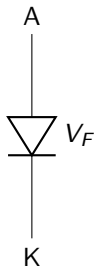
Classification of Power Semiconductor Devices

Device	Terminals	Turn-on	Turn-off
Diode	2	Automatic (forward biased)	Automatic (reverse biased)
DIAC	2	Automatic (breakover voltage, either polarity)	Automatic (current below holding)
SCR	3	Gate pulse (forward biased)	Not gate-controlled – natural/forced commutation
TRIAC	3	Gate pulse (either polarity)	Natural commutation (current crosses zero)
MOSFET	3	Gate voltage	Gate voltage – fully controllable
GaN HEMT ¹	3	Gate voltage	Gate voltage – fully controllable

Uncontrolled (Diode, DIAC) → Semi-controlled (SCR, TRIAC) → Fully-controlled (MOSFET, GaN) – the natural progression this session follows.

¹High Electron Mobility Transistor

Power Diode – Symbol & Principle

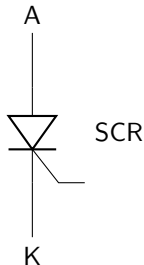


- Two-terminal device: Anode (P-type) and Cathode (N-type), single PN junction.
- **Uncontrolled**: conducts when forward biased ($V_A > V_K$), blocks when reverse biased – no gate terminal at all.
- Switching is decided entirely by circuit conditions, not by any external control signal.
- Key ratings: forward drop V_F , reverse recovery time t_{rr} , peak repetitive reverse voltage V_{RRM} .

Power Diode – Example Parts ☰

Part	Ratings	Notes
MUR1560	600 V, 15 A, $t_{rr} \approx$ 35–60 ns	Ultrafast rectifier – SMPS, inverter freewheel diode
1N5408	1000 V, 3 A, standard recovery	General-purpose bridge/line rectifier

SCR – Symbol & Principle

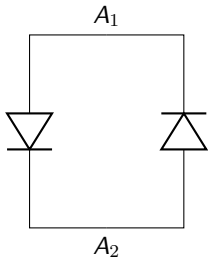


- Four-layer (PNPN) device: Anode, Cathode, and Gate.
- **Semi-controlled:** a gate pulse triggers turn-on when forward biased. Turn-off is *not* gate-controlled – current must fall below the holding current I_H (natural or forced commutation).
- Once latched on, the gate loses all further control.
- Key ratings: $(dv/dt)_{crit}$, I_H , gate trigger current I_{GT} , V_{DRM} .

SCR – Example Parts ≡

Part	Ratings	Notes
BT151	500–800 V, 12 A RMS, $I_{GT} \approx 15$ mA	Medium power, TO-220 – motor/lamp control
2N6509	800 V, 25 A RMS (16 A avg)	Higher current, TO-220 – crowbar, in-rush limiting

DIAC – Symbol & Principle

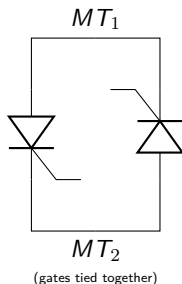


- Two-terminal, three-layer device – *no gate terminal at all*. Behaves like two diodes in anti-parallel, without a gate.
- Symmetric structure: identical behaviour for either current direction.
- Switches to low-impedance conduction only once the breakover voltage V_{BO} is exceeded, in *either* polarity.
- Mainly used as a voltage-triggered pulse source to fire TRIAC/SCR gates – not a power-handling switch itself.

DIAC – Example Parts ☰

Part	Ratings	Notes
DB3	$V_{BO} \approx 28\text{--}36\text{ V}$ (typ. 32 V), $I_{TRM} = 2\text{ A}$	Classic TRIAC-trigger DIAC, DO-35
DB4	$V_{BO} \approx 35\text{--}45\text{ V}$ (typ. 40 V), $I_{TRM} = 2\text{ A}$	Same family, higher breakover option

TRIAC – Symbol & Principle

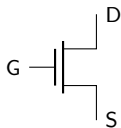


- Functionally two SCRs in **anti-parallel**, sharing a single gate terminal.
- Fully bidirectional: conducts and blocks in both polarities, and can be gate-triggered from either polarity.
- Ideal for direct AC power control – avoids needing a separate anti-parallel pair of SCRs.
- Key ratings: similar to SCR ($I_{T(RMS)}$, V_{DRM} , I_{GT}) but specified for both current directions.

TRIAC – Example Parts ☰

Part	Ratings	Notes
BT136	600 V, 4 A, sensitive gate	Low-power AC switching, dimmers, fan control
BTA16-600B	600 V, 16 A	Snubberless, higher power – heating, motor control

MOSFET – Symbol & Principle



- Voltage-controlled, majority-carrier device: Drain, Source, Gate.
- **Fully controlled:** gate voltage controls *both* turn-on and turn-off – no latching.
- Intrinsic body diode always allows reverse current, regardless of gate state – the MOSFET *cannot* block reverse voltage.
- Fast switching (ns range), but conduction loss ($I^2 R_{DS(on)}$) grows sharply with voltage rating.

MOSFET – Example Parts ☰

Part	Ratings	Notes
IRF450	500 V, 12–13 A, $R_{DS(on)} \approx 0.4 \Omega$	Legacy HEXFET, TO-3 – audio, high-voltage switching
IRF540N	100 V, 33 A, $R_{DS(on)} \approx$ 44 m Ω	Common logic-level MOSFET – motor drive, low-voltage SMPS

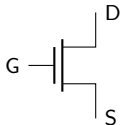
GaN – Why Wide-Bandgap?

Property	Silicon (Si)	Gallium Nitride (GaN)
Bandgap	≈ 1.1 eV	≈ 3.4 eV
Critical E-field	≈ 0.3 MV/cm	≈ 3.3 MV/cm
Electron mobility	Moderate	High (2DEG channel ¹)
Reverse recovery Q_{rr}	Present (body diode)	$\approx$ Zero
Practical switching freq.	100s of kHz	MHz range

A much higher critical field lets GaN use a far thinner drift region for the same blocking voltage – directly translating into lower $R_{DS(on)}$ and faster switching than an equivalent silicon device.

¹two-dimensional electron gas

GaN – Device Structure & Key Features



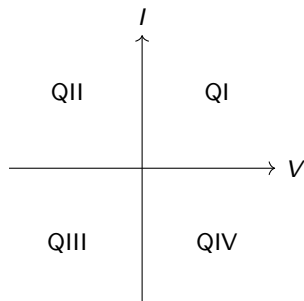
(E-HEMT, drawn as MOSFET-equivalent)

- Enhancement-mode HEMT (E-HEMT): an AlGaIn/GaN heterojunction forms a 2D electron gas (2DEG) channel; gate voltage depletes/restores it.
- Lateral device structure (unlike Si's vertical power MOSFET) – enables very low parasitic inductance packaging.
- Near-zero reverse recovery charge – a major source of switching loss in Si is essentially eliminated.
- Narrower, more sensitive gate voltage window (often 0 V to 6 V) – gate drive design needs extra care.

GaN – Example Parts ☰

Part	Ratings	Notes
GS66508B	650 V, 30 A, $R_{DS(on)} \approx 50 \text{ m}\Omega$	GaN Systems E-HEMT, GaNPX package – HF power converters
EPC2010C	200 V, 22 A, $R_{DS(on)} \approx 25 \text{ m}\Omega$	EPC eGaN FET, die-form – high-speed DC-DC, Class D audio

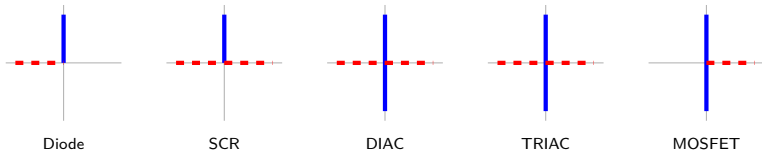
The V-I Quadrant Plane



- Every switch's behaviour can be plotted on the V - I plane of the device itself.
- A device **conducts** along the I -axis (small V , current flowing); it **blocks** along the V -axis (small I , voltage held off).
- Classifying a device this way – which quadrant(s) it can conduct in, and which it can block in – is the basis of *switch synthesis*: building the switch a converter needs from simpler devices.

Approach following Prof. G. Narayanan, IISc Bangalore.

Quadrant Occupancy of Each Device



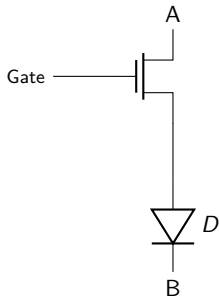
Solid = conduction (I -axis) Dashed = blocking (V -axis)

DIAC and TRIAC look identical here – both conduct and block in *both* directions. The difference is control: DIAC switches on voltage alone; TRIAC needs a gate pulse. MOSFET conducts both ways when ON but blocks only one polarity when OFF – the body diode always conducts in reverse.

Why Switch Synthesis Matters

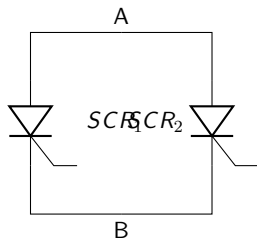
- Real converter legs often need a switch with quadrant capability that *no single device* provides on its own.
- Example: a fully voltage- and current-bidirectional (four-quadrant) switch, needed in matrix converters and some AC drives, does not exist as a single silicon part outside the TRIAC's specific case.
- The answer is to **combine** simpler devices – in series, in parallel, or anti-parallel – to synthesize the switch the application actually needs.
- The next three slides cover each combination and what it buys you.

Synthesis: Series Combination



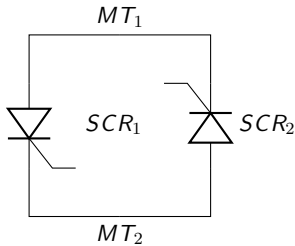
- Stacking two devices in the *same* current path adds their blocking capability.
- A MOSFET alone cannot block reverse voltage (body diode always conducts). Adding a series diode *D* blocks the direction the MOSFET cannot.
- Trade-off: current can now flow in only one direction through the pair, even though the MOSFET itself could conduct both ways when ON.

Synthesis: Parallel Combination =



- Connecting two *identical* devices between the same pair of nodes shares the current between them.
- Effect: raises the total current-carrying capacity beyond a single device's rating.
- Needs matched device characteristics (or small series-sharing resistors/inductors) in practice, or one device can hog the current and overheat.

Synthesis: Anti-Parallel Combination ↻



- Connecting two devices back-to-back (reversed polarity) between the same two nodes lets current flow in *either* direction.
- This is exactly how a TRIAC is built internally: two SCRs anti-parallel, gates tied together.
- Same principle explains the MOSFET's own bidirectional current capability – its intrinsic body diode is effectively a built-in anti-parallel diode.
- Other common case: SCR anti-parallel with a diode – SCR handles the controlled direction, diode freewheels the other.

Comparison: All Devices at a Glance

Device	Typ. V range	Typ. I range	Switching speed	Typical Application
Diode	up to kV	up to 100s A	Fast (ns– μ s recovery)	Rectification, freewheeling
SCR	up to kV	up to 1000s A	Slow (commutation-limited)	AC power control, HVDC
DIAC	~ 30 V	mA range	N/A – trigger device	Triggering TRIACs/SCRs
TRIAC	up to ~ 1 kV	up to 100s A	Slow (line frequency)	AC lamp/motor/heater control
MOSFET	up to ~ 1 kV (Si)	up to 100s A	Fast (ns– μ s)	SMPS, motor drives
GaN HEMT	up to ~ 650 V (typ.)	up to 100s A	Very fast (MHz range)	HF SMPS, EV chargers

- The progression from diode → SCR/TRIAC → MOSFET/GaN is a progression in *control*: none, turn-on only, then full turn-on/turn-off control.
- DIAC and TRIAC are naturally bidirectional; diode, SCR, and MOSFET are not – and each has a different reason why.
- GaN's advantage over silicon comes directly from its wider bandgap and higher critical field, not from a fundamentally different operating principle.
- Quadrant thinking is the practical bridge from “what a device can do” to “what a converter leg needs” – series, parallel, and anti-parallel combinations synthesize the gap.
- Always re-verify part ratings against the current datasheet before a design commitment; values here are for teaching reference.

References: M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*; N. Mohan, T. M. Undeland, W. P. Robbins, *Power Electronics: Converters, Applications and Design*; switch-synthesis / quadrant-switch treatment following Prof. G. Narayanan, IISc Bangalore; manufacturer datasheets (ON Semiconductor, NXP, STMicroelectronics, Infineon, GaN Systems, EPC).